

Quiz 1 (Version 2)

CAS CS 132: *Geometric Algorithms*

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- ▷ You will have approximately 35 minutes to complete this exam.
- ▷ Your final solution must appear in the solution boxes for each problem. **Only include your final solution in the solution box. You must show your work outside of the solution box.** You will not receive credit if you don't show your work.

1 Row Operations

Apply the row operations:

$$R_3 \leftarrow R_3 + R_1$$

$$R_2 \leftarrow 3R_2$$

$$R_1 \leftrightarrow R_3$$

from top to bottom to the following matrix. You must write down the intermediate matrices and row operations you used in your calculation.

$$\begin{bmatrix} 0 & -2 & 8 \\ -10 & -4 & -2 \\ 3 & 6 & -5 \end{bmatrix} \xrightarrow{R_3 \leftarrow R_3 + R_1}$$

$$\begin{bmatrix} 0 & -2 & 8 \\ -10 & -4 & -2 \\ 3 & 4 & 3 \end{bmatrix} \xrightarrow{R_2 \leftarrow 3R_2}$$

$$\begin{bmatrix} 0 & -2 & 8 \\ -30 & -12 & -6 \\ 3 & 4 & 3 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_3}$$

Solution.

$$\begin{bmatrix} 3 & 4 & 3 \\ -30 & -12 & -6 \\ 0 & -2 & 8 \end{bmatrix}$$

2 Reduced Echelon Forms

Determine the reduced echelon form of the following matrix. You must write down the intermediate matrices and row operations you used in your calculation.

$$\begin{array}{c}
 \begin{bmatrix} -1 & -2 & -1 & 7 \\ -3 & -6 & -6 & 27 \\ -2 & -4 & 1 & 8 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2 - 3R_1} \begin{bmatrix} -1 & -2 & -1 & 7 \\ 0 & 0 & -3 & 6 \\ 0 & 0 & 3 & -6 \end{bmatrix} \xrightarrow{R_3 \leftarrow R_3 + R_2} \begin{bmatrix} -1 & -2 & -1 & 7 \\ 0 & 0 & -3 & 6 \\ 0 & 0 & 0 & 0 \end{bmatrix} \\
 \begin{bmatrix} -1 & -2 & -1 & 7 \\ 0 & 0 & -3 & 6 \\ -2 & -4 & 1 & 8 \end{bmatrix} \xrightarrow{R_3 \leftarrow R_3 - 2R_1} \begin{bmatrix} -1 & -2 & -1 & 7 \\ 0 & 0 & -3 & 6 \\ 0 & 0 & 3 & -6 \end{bmatrix} \xrightarrow{R_3 \leftarrow R_3 + R_2} \begin{bmatrix} -1 & -2 & -1 & 7 \\ 0 & 0 & -3 & 6 \\ 0 & 0 & 0 & 0 \end{bmatrix} \\
 \xrightarrow{R_2 \leftarrow R_2 / -3} \begin{bmatrix} -1 & -2 & -1 & 7 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \leftarrow R_1 + R_2} \begin{bmatrix} -1 & -2 & 0 & 5 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \\
 \xrightarrow{R_1 \leftarrow -R_1} \begin{bmatrix} 1 & 2 & 0 & -5 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}
 \end{array}$$

Solution.

$$\begin{bmatrix} 1 & 2 & 0 & -5 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

3 General Form Solutions

Determine a general form solution for a linear system whose augmented matrix is row equivalent to the following matrix (*Hint*. First determine if the following matrix is in reduced echelon form).

$$\begin{bmatrix} 1 & 0 & -4 & 0 & 5 & 5 \\ 0 & 1 & 2 & 1 & 0 & 6 \\ 0 & 0 & 0 & 1 & 6 & 6 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & -4 & 0 & 5 & 5 \\ 0 & 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 6 & 6 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Solution.

$$x_1 = 5 + 4x_3 - 5x_5$$

$$x_2 = -2x_3$$

x_3 is free

$$x_4 = 6 - 6x_5$$

x_5 is free